W-H-196J
December 14, 1973
SUPERSEDING
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June 18, 1971 and
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September 17, 1971

FEDERAL SPECIFICATION

HEATER, WATER, ELECTRIC, AND GAS FIRED, RESIDENTIAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric and gas fired automatic water heaters with a storage tank and all necessary controls and accessories mounted as an integral unit on a base to make a complete, self-contained domestic water heater.

1.2 Classification. Water heaters shall be of the applicable group, series, type, size, and model as specified (see 6.2 and 6.3).

Group A - Electric water heaters.
  Series 1 - Single heating unit.
  Series 2 - Double heating unit.

Group B - Gas fired water heaters.

Type II - Cement lined steel tank.
Type III - Copper-silicon tank.
Type IV - Glass-lined tank.
Type VI - Copper-lined tank.
Type VII - Nickel-copper alloy tank.
Type VIII - Copper tank.
Type IX - Stainless steel tank.
Type X - Copper-nickel alloy tank (90 percent Cu, 10 percent Ni).

Size and model shall be in accordance with table I, II, or III as applicable.

TABLE I. Group A-electric, series 1-single heating units

Nominal size (Capacity gallons ±10%)	Model	Power input regular (watts)1/	Power input speed heater (watts)1/
20	Round	1,500	2,500
20	Table top	1,500	2,500
30	Round	1,500	4,500
30	Table top	1,500	4,500
40	Round	2,000	4,500
40	Table top	2,000	4,500
50	Table top	2,500	4,500
52	Round	2,500	4,500
66	Round	3,000	4,500
80	Round	4,000	4,500
100	Round	---	4,500
120	Round	---	4,500

1/ Power inputs different than those listed in table I and table II are available and procurement officers should consult local utility companies for the desired power input for their particular area. Power inputs specified shall conform to tolerances specified in UL 174.

FSC 4520

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TABLE II. Group A-electric, series 2-double heating units

Nominal size (Capacity gals. ±10%)	Model	Power input, regular(watts)1/ Lower unit	Upper unit	Power input, speed heater(watts)1/ Lower unit	Upper unit
20	Table top	600	1,000	2,500	2,500
30	Table top	600	1,000	4,500	4,500
30	Round	600	1,000	4,500	4,500
40	Round	750	1,250	4,500	4,500
40	Table top	750	1,250	4,500	4,500
52	Round	1,000	1,500	4,500	4,500
50	Table top	1,000	1,500	4,500	4,500
66	Round	1,250	2,000	4,500	4,500
80	Round	1,500	2,500	4,500	4,500
100	Round	---	---	4,500	4,500
120	Round	---	---	4,500	4,500

1/ Power inputs different than those listed in table I and table II are available and procurement officers should consult local utility companies for the desired power input for their particular area. Power inputs specified shall conform to tolerances specified in UL 174.

TABLE III. Group B-gas fired, minimum normal recovery capacity

Tank size	Recovery capacity
(Gallons)	(Gallons per hour @100°F(37.8°C)-rise)
20	18.5 or 22.7, as specified
30	25.2 or 33, as specified
40	27.7 or 35, as specified
50	30 or 42, as specified
60	35
65	63
75	42 or 63, as specified
85	63
100	63

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-576 - Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet, and Strip).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys; Rod, Wire, Shapes, Forgings, and Flat Products, (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-M-31 - Magnesium Alloy, Bars, Rods, and Special Shaped Sections, Extruded.
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion-Resisting.
SS-C-192 - Cement, Portland.
PPP-P-600 - Porcelain Enamel Products and Household Appliances, Electrical and Mechanical-Requirements for Packaging and Packing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Goverment for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-V-3 - Valves, Fittings, and Flanges; Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-S-8840 - Steel Sheet and Strip, Corrosion-Resistant, Precipitation Hardening (AM350 and AM355), Premium Quality.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific isssue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute (ANSI) Standards:

Z21.10.1 - Gas Water Heaters.
Z21.15 - Manually Operated Gas Valves.
Z21.18 - Domestic Gas Appliance Pressure Regulators.
Z21.22 - Relief Valves and Automatic Gas Shutoff Devices for Hot Water Supply Systems.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10016.)

American Iron and Steel Institute:

Steel Products Manual - AISI Sheet Steel.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

American Society for Testing and Materials (ASTM) Standard:

A240 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Steet, Philadelphia, PA 19103.)

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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Underwriters' Laboratories Inc. (UL) Publication:

UL 174 - Standard for Electric Storage Tank Water Heaters.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, NY 11749; 207 E. Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), the contractor shall furnish within the time limit specified in the contract, one water heater and one tank of each group specified in the contract as preproduction samples for examination and testing. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government. Approval of the preproduction samples on the basis of the specified examination and tests shall not relieve the contractor of his obligation to meet the requirements of this specification for the remaining units furnished under the contract.

3.2 Standard product. The water heaters delivered under this specification shall be the manufacturer's commercially current standard product, except for any changes necessary to comply with the requirements of this specification. All water heaters of one group, series, type, size, and model furnished under this specification for any one contract, including parts and assemblies thereof, shall be interchangeable. Replacement and adjustment of burner or heating elements, controls and relief valves shall be accomplished by removal of access panels, but not require removal or cause injury to major parts such as insulation, top pan, jacket, and water piping. Once set, service adjustments shall not be required to controls unless a change is required in heater performance. Accessible adjustments shall be returnable to their original settings.

3.3 Standard compliance. Except as otherwise specified herein, electric and gas fired water heaters shall conform to the applicable requirements of UL Std. 174 and ANSI Stds. Z21.10.1.

3.3.1 Labels and listings.

3.3.1.1 Underwriters' Laboratories, Inc. Standards (UL). Each contractor shall submit to the contracting agency proof that the electric water heaters he proposes to supply under this specification conform to the requirements of UL Std. No. 174 for electric storage tank water heaters. The label, or listing with re-examination, of the UL, may be accepted as evidence that the electric water heaters conform to this requirement. In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the electric water heaters conform to the applicable requirements of the published standards, including methods of tests of the UL Std. No. 174.

3.3.1.2 American Gas Association Laboratories (AGA). Each contractor shall submit to the contracting agency proof that the gas fired water heaters he proposes to supply under this specification conform to the requirements of ANSI Std. Z21.10.1 for gas water heaters. The label, of the AGA, may be accepted as evidence that the gas water heaters conform to this requirement. In lieu of the label or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the gas fired water heaters conform to the applicable requirements of the published standards, including methods of test of the ANSI Std. Z21.10.1.

3.4 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.5 Splashboard. Unless otherwise specified (see 6.2), tabletop heaters may be furnished with or without a splashboard at the option of the manufacturer. When specified, the splashboard shall be porcelain coated to match the tabletop or other finish when specified (see 6.2).

3.6 Tanks. Heater tanks shall be manufactured to the following materials requirements, as appropriate for the type of heater specified (see 1.2 and 6.2). The minimum working pressure of the tank shall be 150 pounds per square inch gage pressure (p.s.i.g.).

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3.6.1 Cement-lined (stone-lined) steel tanks, group A, type II. The lining of a cement lined tank shall be a low-soluble hydraulic-cement lining material containing not more than 35 percent calcium oxide, and not less than 25 percent silicon. Cement for the lining shall conform to SS-C-192, type as best suited for the purpose. The lining shall be properly cured and completely cover the interior of the tank. The joints at the tapping shall be constructed to prevent corrosion in back of the lining. The lining shall be applied so as to provide a thickness of 0.375 inch plus or minus 0.062 inch and shall shall have the same coefficient of expansion as the shell (see 4.4.1.1). The water absorption of the lining material shall be not more than 17 percent of the dry weight when tested in accordance with 4.4.1.2.

3.6.2 Type III, copper-silicon tank. Copper-silicon alloy shall conform to QQ-C-591, alloy No. 653 or 655, soft temper.

3.6.3 Type IV, glass-lined steel tank (porcelain enamel coating) (see 6.8).

3.6.3.1 Tank steel. Steel for fabrication of water storage tanks shall be of a type recommended in the AISI Steel Products Manual as suitable for exposure to fire and the products of combustion to which the tank will be exposed.

3.6.3.2 Glass lining. The glass lining shall be applied to all interior surfaces that are exposed to hot water including manhole and manhole covers when applicable. The glass lining shall be fused to the steel base by firing at a temperature of not less than 1500°F (815.5°C) and shall conform to the requirements of 3.6.3.2.a through 3.6.3.2.d for coverage, 3.6.3.2.e for solubility, and 3.6.3.2.f for chemical analysis.

a. Thickness. The thickness of the glass lining shall be not less than 0.006 inch or greater than 0.020 inch.

b. Discontinuities. Discontinuities shall not be more than 0.025 square inches per square foot of internal surface excluding, edges, fittings and welded areas after applying the glass lining. No single defect shall be greater than 0.125 inch in diameter.

c. Edges and fittings. All edges and fittings shall be coated with glass lining except where sharp edges cannot be avoided a slight "burn off" (see 6.6), of the lining that does not extend more than 0.062 inch back from the sharp edge shall be allowed. All fittings that are welded to the tank shall be coated with the lining on those areas that are exposed to the water, except for threads and small areas immediately adjacent to them.

d. Welds. Removal of the glass lining due to welding or mating of parts after glass lining shall not result in exposed base metal in excess of 0.025 square inches per gallon of storage capacity.

e. Glass lining solubility. The weight loss of the glass lining shall not exceed 15 milligrams per square inch when tested as specified in 4.4.2.3.

f. Chemical analysis. Prior to discarding the first test cycle solution obtained in accordance with 4.4.2.3.g.(4), the solution shall be tested as specified in 4.4.2.4. The solution shall not contain more than trace amounts of lead, arsenic, or antimony.

3.6.4 Type VI copper-lined steel tank. Copper linings shall be electrolytically or mechanically bonded to the steel tank shell. The lining shall cover all interior portions of the tank exposed to hot water. The thickness of the copper lining shall be not less than 0.023 inch. Material used for the lining shall contain not less than 89 percent copper and shall conform to QQ-C-576, cold rolled, light temper, or QQ-C-591, copper alloy No. 653 or 655, soft temper.

3.6.5 Type VII nickel-copper alloy tank. Nickel-copper alloy shall conform to QQ-N-281, class A, form 4, cold rolled and annealed (plain finish).

3.6.6 Type VIII copper tank. Copper shall conform to QQ-C-576, cold rolled, light temper, tensile strength 38,500 p.s.i. minimum.

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3.6.7 Type IX stainless-steel tank. Stainless steel shall conform to QQ-S-766, class 304, 304L, 316 or 316L, condition A, or MIL-S-8840, composition AM350, annealed condition, or ASTM, A240, type XM8.

3.6.8 Type X copper-nickel alloy tank. Copper-nickel alloy shall conform to MIL-C-15726, composition 90-10, temper soft.

3.7 Cathodic protection. Unless otherwise specified (see 6.2), magnesium anodes shall be provided for cathodic protection for glass lined type IV, tank or heater. The anode shall be located so as to insure protection to all interior parts of the vessel. An electrical connection shall be established and maintained between the anode and the metal tank (heater) shell.

3.7.1 Magnesium anode. The magnesium anode shall be fabricated from material conforming to QQ-M-31, composition number AZ31B with the following exceptions:

Material	Maximum percent
Copper	0.04
Nickel	0.001
Iron	0.002

Shapes of anodes shall be as applicable and anodes shall be reinforced with steel wire not less than 0.091 inch in diameter. The weight of the anode in each tank shall be not less than 25 grams per square foot of surface within the tank shell.

3.7.2 Other anodes. When anodes other than magnesium are used, the tank (heater) wall potential shall not be more positive than minus 0.85 volts when tested as specified in 4.4.3.

3.8 Bottom cold water inlet. When the cold water inlet is located at the bottom of the tank, an anti-siphon valve (vacuum breaker) complete with installation instructions shall be furnished with each heater. Check valves are not acceptable as anti-siphon valves.

3.9 Tappings and fittings. Tappings and fittings for cement and glass-lined tanks shall be carbon steel, copper-lined steel, brass, tin or brass plated steel, cooper silicon, or copper plated steel having internal waterways lined with electrically inert materials. For copper-silicon tanks, the tappings and fittings shall be the same material as the tank and shall conform to QQ-C-591. For copper-nickel, nickel-copper, and stainless-steel tanks the tappings and fittings shall be stainless steel or the same material as the tank. Threads for tappings and fittings shall be in accordance with Handbook H28. Unless otherwise specified (see 6.2), the inlet, outlet, and drain connections shall be 3/4 inch NPT minimum size.

3.10 Safety. All parts that are subject to high operating temperature shall be insulated, enclosed, or guarded where exposed to contact by personnel or a fire hazard is created. All moving parts that are of such nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions.

3.11 Jacket finish. Unless otherwise specified (see 6.2), the exposed finish of the exterior of the water-heater jacket shall be organic enamel. Tabletop heaters shall be equipped with a white porcelain top unless otherwise specified.

3.12 Relief valve. Unless otherwise specified (see 6.2), a combination temperature, pressure relief valve conforming to ANSI Z21.22 shall be installed on or supplied with the water heater. If the valve is not installed, complete instructions for installation shall be supplied with the valve. The pressure setting of valve shall in no case exceed the working pressure of the heater tank being supplied (see 6.1.2).

3.13 Temperature control, Group A - Electric water heaters. Thermostats shall be provided in accordance with UL 174 except that when two thermostats are furnished, the thermostat controlling the upper heating element shall be set at 140°F(60.0°C) and the thermostat controlling the lower heating element shall be set at 150°F(65.6°C).

3.13.1 Interlocking device. Electric water heaters with upper and lower heating elements shall be equipped with an interlocking device (block switch) which shall prevent the two heating elements from operating simultaneously.

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3.14 Operating current, Group A - electric water heaters. Heating units shall operate on alternating or direct current as specified (see 6.2).

3.15 Voltage. Alternating current water heaters shall have nameplate voltage ratings of 230-240 volts and shall be suitable for operation on 208 volts network systems (see 6.1.1). Direct current water heaters shall be suitable for operation on 210 to 250 volts.

3.16 Recovery capacity group B water heaters. The normal recovery capacity shall be not less than that shown in table III and shall be based on a temperature rise of 100°F. with cold water supplied at 70°F. plus or minus 2°F.

3.17 Repair parts. When specified (see 6.2), repair parts shall be furnished with each heater in the quantity specified and shall be packed and shipped with the heater.

3.18 Marking. When specified (see 6.2), heaters shall be marked in accordance with MIL-STD-130 in addition to the markings required in 3.3. Information to be marked shall be as specified (see 6.2).

3.19 Workmanship. The general construction and assembly of the water heater shall be of a neat and workmanlike character. The water heaters shall be free of sharp edges, scratches on enameled surfaces, loose edges, dents or bent on the outer jacket, or any other defect affecting the appearance or serviceability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Component certification. In lieu of using his own facilities, the contractor may submit a report of analysis and performance of component materials from a laboratory acceptable to the government, whether an independent testing facility or the laboratory of the component material supplier, as evidence of compliance with the specification requirements. The report shall include, in additon to the actual data, the manufacturer's name, name and address of the testing laboratory, and date of test.

4.2 Visual examination. (see 6.4).

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all water heaters of the same group offered for delivery at the same time.

4.2.2 Sampling for examination of end item. Sampling for examination shall be in accordance with level S-2 of MIL-STD-105.

4.2.3 End item examination. Water heaters shall be examined for defects listed in table IV. The acceptable quality level (AQL) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

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TABLE IV. Classification of defects - end item

Examine	Defect	Classification	
		Major	Minor
Unit	Group, series, type, size, model, power input, or recovery capacity not as specified (see 1.2).....	X	
Interchangeability and maintainability	Not as specified (see 3.2).....	X	
Identification and instructional data plates.	Missing, illegible, not properly located, or not complete (see 3.3).....		X
Materials.	Not as specified (see 3.4 and 3.6).....	X	
Splashboard.	Not as specified (see 3.5).....		X
Cathodic protection.	Anode not as specified (see 3.7).....	X	
Anti-siphon valve.	Missing or not as specified (see 3.8).....	X	
Tappings and fittings.	Not as specified (see 3.9).....	X	
Exposed parts.	Not as specified (see 3.10).....	X	
Jacket finish.	Not as specified (see 3.11).....		X
Detached items.	Written instructions not furnished for installation when applicable (see 3.12)..		X
Operating current.	Not as specified (see 3.14).....	X	
Voltage.	Not as specified (see 3.15).....	X	
Repair parts.	Not as specified (see 3.17).....		X
Marking.	Illegible or not as specified (see 3.18)..		X
Workmanship.	Components missing or broken.....	X	
	Moving parts stick or bind.....	X	
	Sharp edges hazardous to operating personnel.....	X	
	Loose screws, dents, or bent on outer jacket.....		X
	Damage and defects affecting function and serviceability.....	X	
	Damage and defects not affecting function and serviceability.....		X

4.3 Tests.

4.3.1 Preproduction tests. Preproduction tests shall consist of the applicable tests in accordance with 3.3, 4.4.1 or 4.4.2, and 4.4.3.

4.3.2 Frequency of tests. Unless otherwise specified, (see 6.2), tests shall be performed on the preproduction samples and thereafter only when changes in material or design are made (see 6.5).

4.4 Test methods.

4.4.1 Cement lining tests.

4.4.1.1 Thickness test. The thickness of the lining shall be determined by spear measurements with a hardened steel point not larger than 0.062 inch in diameter. At least three measurements shall be made, including one each on bottom head, topnead, and shell. The measurements shall be taken immediately after the lining is applied and before the cement has set. When specified (see 6.4), the manufacturer shall supply certification of compliance with the thickness requirements of 3.6.1 with the data upon which certification is based.

4.4.1.2 Absorption test.

a. Samples. Six test specimens approximately 6 inches square of clean, dry, cement lining identical in mix, thickness, method of application, and curing process to the lining in the tanks concerned shall be supplied.

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b. Procedure. The water absorption of the cement lining material shall be determined by immersing dry specimens (see 4.4.1.2.a) of the tank materials in water and comparing the weights of the specimens before and after immersion. The weight of each specimen shall be determined by the manufacturer to the nearest 0.01 gram before the lining is applied. The dry weight of the lining material shall be determined as follows: Heat the lined test specimen at a temperature of not less than 230°F(110°C) for a period of not less than two hours; cool to approximately room temperature and weigh. The difference between the weights of the corresponding unlined specimens (weights supplied by the manufacturer), and of their weights after heating is the dry weight of the lining material. The water absorption shall be determined in the following manner: Immediately after the heating, cooling, and weighing, immerse the specimens for a period of five hours in distilled water maintained at a temperature of 180°F(82.2°C); transfer to a bath of distilled water at room temperature and allow the specimens to cool to hand handling temperature; remove the specimens; wipe them dry, and weigh immediately to the nearest 0.01 gram. The difference between the weights of the specimens before and after immersion is the weight of the water absorbed. The absorption shall be reported as a percentage of the dry weight of the lining material. The mean value of at least two specimens shall be used to determine compliance with 3.6.1. Nonconformance shall consitute failure of this test.

4.4.2 Glass lining tests.

4.4.2.1 Lining coverage. A glass lined tank that has not been subjected to test pressure greater than 150 p.s.i.g. shall be cut into four or more segments using a band saw, bayonet saw, or other suitable mechanical cutting device. Each segment shall be visually inspected for the presence of areas of exposed metals except that cracked and spalled areas of lining caused by the cutting operation shall not be considered. Exposed areas of metal shall be measured with a low powered microscope with a calibrated eyepiece. Nonconformance to 3.6.3.2.b shall constitute failure of this test.

4.4.2.2 Thickness of glass lining. Thickness measurements shall be taken at not less than 10 representative points on the shell and heads of the tank using a calibrated magnetic or similar thickness gauge. Nonconformance to 3.6.3.2.a shall constitute failure of this test.

4.4.2.3 Solubility of glass lining.

a. Apparatus. The apparatus illustrated in figure 1 shall be used for determining the solubility of the glass lining. The cylinder assembly shall be constructed of 16 gage type 304 stainless steel; the ends shall be ground so as to conform with the curvature of the tank to be tested. The gaskets shall be made of neoprene rubber. The cell shall be heated with a small gas burner or a small electric heater placed immediately below the cylinder assembly midway between the ends.

b. Test specimens. The test specimens for this test shall consist of four 3-1/2 inch by 3-1/2 inch sections cut from the outer wall (not the center flue) of the tank approximately midway between the ends. These specimens may be taken from the center segments cut in 4.4.2.2.

c. Preparation of specimens. The specimens shall be buffed with a rubbing stone or suitable abrasive paper around the entire perimeter of the panel to completely remove rough edges and glass lining fragments. The specimens shall then be scrubbed on both sides using a nylon brush and a mildly abrasive detergent cleaner powder, rinsed with distilled water, dried for 1 hour in a drying over at 225°F(107°C) and placed in a desiccator while hot.

d. Initial weight measurements. The specimens, after cooling to room temperature, shall be weighed to the closet 1/10 milligram.

e. Calibration of test cell. Because of small variations in cell dimensions and configuration, each test cell must be calibrated before it is used. The procedure is as follows:

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(1) Assemble the cell as it is to be operated.
(2) Fill with water to just below where the condenser tube is welded into the cylinder (see figure 1).
(3) Adjust the input of the heat source to give a slow, rolling boil. If water rises in the condenser, remove small amounts of water until the cell will operate without surging.
(4) Remove from the heat and check to see if water level is completely covering the panels.
(5) Cool to room temperature, then measure the volume of water contained in the cell.
(6) Record this volume in the cell and use this amount of test solution in all subsequent tests.

f. Test solution. The test solution used for each exposure cycle shall consist of 400 milligrams of reagent grade sodium bicarbonate dissolved in 1 liter of distilled water.

g. Test procedure.

(1) Assemble the test cell using weighed specimens from the same tank on each end of the test cell.
(2) Pour the correct volume of test solution into the cell.
(3) Adjust the input of the heater to give a consistent slow, rolling boil.
(4) After 18 hours boil time, dismantle the cell and discard the used test solution (see 3.6.3.2.f and 4.4.2.4).
(5) Clean the specimens using a soft cloth wetted with a solution of 1 percent trisodium phosphate to remove loosly-adhearing deposits, wipe dry and store in desiccator between solution treatments.
(6) After 8 cycles of 18 hours each, clean specimens as above, rinse with distilled water and dry at 225°F(107°C) for 1 hour. Place the specimens in a desiccator while hot and after cooling to room temperature weigh to the closet 1/10 milligram.

h. Calculation of results. The exposed area of each specimen shall be calculated from the average diameter of the test spot, measured to the closest 0.015 inch (3.96 mm) taken in three different locations. The solubility of the lining shall be calculated by dividing the weight loss in milligrams by the exposed area in square inches.

i. Interpretation of results. Nonconformance with 3.6.3.2.e shall constitute failure of this test.

4.4.2.4 Chemical analysis. The solution obtained in accordance with 4.4.2.3.g.(4) shall be tested by recognized chemical procedures for compliance with 3.6.3.2.f. Chemical tests are not required when it can be determined that no compounds of lead, arsenic, or antimony were used in producing the glass lining material.

4.4.3 Cathodic protection test. The heater tank shall be filled with water having a specific conductance in the range of 400 to 600 micromhos (2500 to 1667 ohm-cm). The tank wall potential shall be measured in reference to a standard copper sulphate cell located inside the tank adjacent to but not in contact with the tank shell and at a maximum distance from the anode or anodes. Measure the tank wall potential with a portable potentiometer or high-resistance volt meter (see 3.7.2).

4.5 Inspection of preparation for delivery requirements. The preservation, packaging and packing shall be examined and tested for compliance with preparation for delivery requirements of section 5. When a manufacturer's packing has met the packing test requirements and certified test records are maintained for a particular model heater and design packing, the manufacturer will not be required to retest unless he changes heater design, packing or has not furnished materials to the government under this specification for a period of six months or more.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging shall be level A, B, or C. Unless otherwise specified in the contract or order packaging shall be level C (see 6.2).

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5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and that would otherwise increase cubage or in accordance with the recommendations of the equipment manufacturer.

5.1.1.2 Preservative application. Unpainted or uncoated surfaces subject to corrosion shall be coated with the applicable type preservative of MIL-P-116 which is designed for coating the material used.

5.1.1.3 Valves and fittings. Preservation and packaging of valves and fitting not installed on the equipment and those removed during disassembly shall be in accordance with level A requirements of MIL-V-3.

5.1.2 Level B, Civil agencies. The heaters shall be packaged as specified in 5.1.1, except preservative shall not be applied.

5.1.3 Level C. The heaters shall be packaged in a manner that will afford protection against mechanical and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his own commercial pack providing it fulfills these requriements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2). When no level is specified in the contract or order, packing shall be Level C.

5.2.1 Level A, B or C. Heaters shall be packed in accordance with PPP-P-600, except level A packing, for military activities fiberboard containers shall not be permitted and in Level B packing fiberboard containers shall be weather-resistant.

5.2.2 Marking. Marking shall be in accordance with Federal Specification PPP-P-600.

6. NOTES

6.1 Intended use. The water heaters are intended to provide hot water for residential or similar systems.

6.1.1 Voltage. The alternating current water heaters covered by this specification, of design voltage 236 volts, are suitable for use on 208 volt systems. However, when heaters with a design voltage of 236 volts are used on a 208 volt circuit, their heating and recovery rate will only be approximately 75 percent of that on a 240 volt circuit. When heaters of design voltage 208 volts or 115 volts are desired, the invitation for bids shall so state, specifically (see 6.2).

6.1.2 Relief valves. It is essential that tanks be protected against excess pressures to prevent ruptures and explosions and that the temperature of the heated water be limited to 210°F(98.9°C) to prevent discharge of live steam. It is recommended that contracts require that relief valves be furnished with each heater, thereby assuring a good quality valve, proper rating (see 3.12), and installation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Group, series, type, size, model, power input, and recovery rate as applicable (see 1.2).
(c) When preproduction samples are required (see 3.1).
(d) If tabletop models are to be furnished with splashboards (see 3.5).
(e) Tank material required (see 3.6).
(f) When magnesium anodes are not required (see 3.7).
(g) If tappings and fittings are to be any size other than 3/4 inch (see 3.9).
(h) If finish other than white enamel is required (see 3.11 and 3.5).
(i) If a relief valve is not to be furnished (see 3.12).
(j) Whether heater is for use with direct or alternating current (see 3.14).
(k) Repair parts required (see 3.17).
(l) Markings required (see 3.18).
(m) Size of lot, if different from 4.2.1.
(n) Frequency of tests, if different (see 4.3.2).
(o) Level of preservation, packaging, and packing required (see 5.1 and 5.2).
(p) When heaters of design voltage 208 volts or 115 volts are desired (see 6.1.1).
(q) When certification is required (see 6.4).

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6.3 Type II, cement lined tanks. Type II, cement lined tanks are available in group A-electric waters heaters only.

6.4 Certification. A certification, signed by a responsible person of the contracting firm, may be required by the purchasing officer listed in the invitation for bids, certifying that the heaters supplied comply with the requirements of this specification (see 6.2).

6.5 Preproduction model. Any changes or deviations of production heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish heaters conforming to this specification.

6.6 Burn off. Burn off is a term used to designate a condition caused by saturation of the lining material layer by iron oxide during the firing operation. The condition usually occurs on areas where the glass lining (porcelain enamel) has been applied very thin. It can be recognized by (a) a complete lack of glossiness, and (b) a reddish brown or black appearance of the lining over the affected area.

6.7 Packaging. Level B packing requirements have been developed by GSA for civil agency procurements. For Navy procurements, details of level B packaging, when required, should be developed at time of procurement when all conditions are known.

6.8 Terminology. The terms "glass lined" and "porcelain enameled" are used interchangeably.

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review activities:

Army - GL, MD

User interest:

Army - CE
Navy - MC

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
HEW-FCC
USDA-ARS

Military Coordinating Activity:

Army-ME

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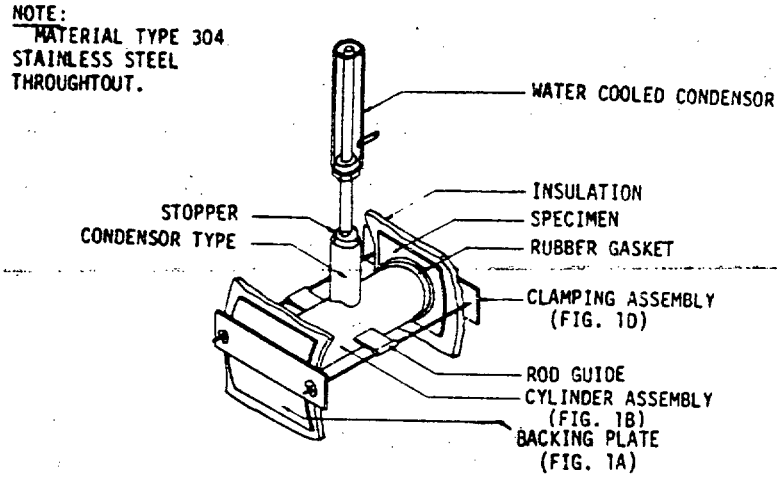


FIGURE 1. Test cell solubility of enamel.
(See also figs. 1A to 1D)

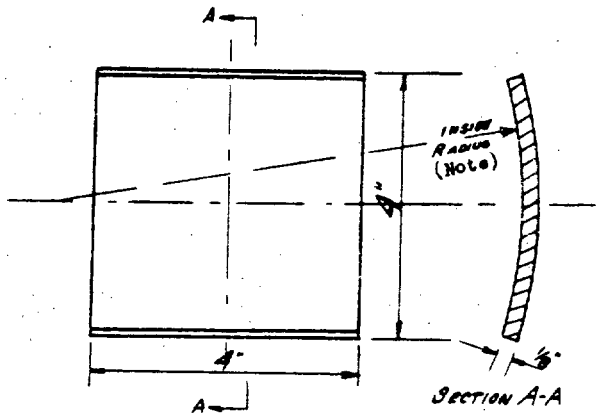


FIGURE 1A. Backing plate. (2 required)
Note: The radius shall be the same as the inside radius of the tank from which specimens are cut.

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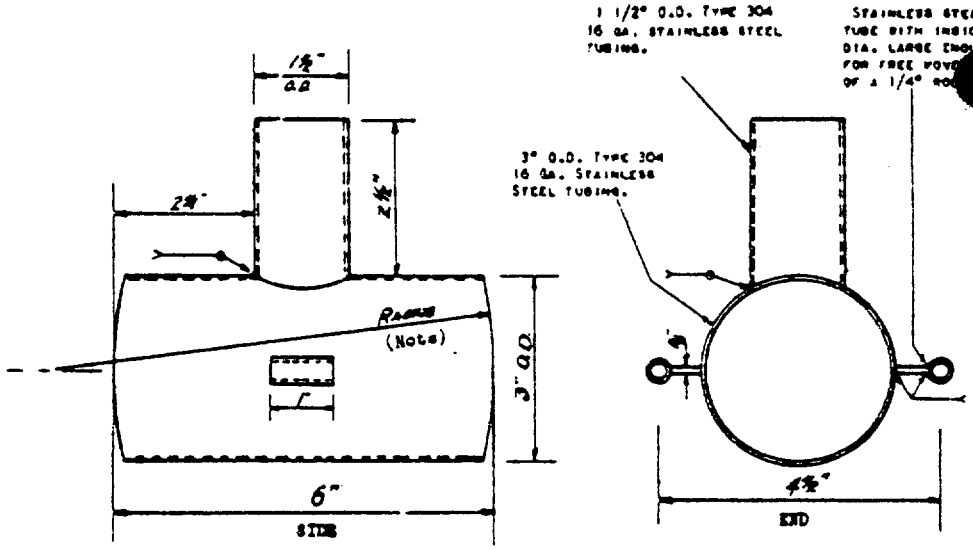


FIGURE 1B. Cylinder assembly. (1 required)
Note: The radius shall be the same as the inside radius of the tank from which specimens are cut.

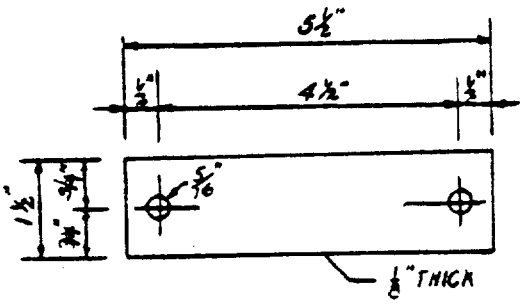


FIGURE 1C. Clamping strip. (2 required)

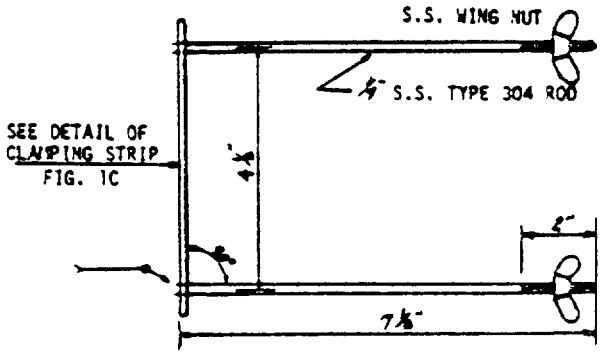


FIGURE 1D. Clamping assembly. (1 required)

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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